COMP2711 / DISCRETE MATHEMATICAL TOOLS

Probability

Past Paper Solutions

59	3	2015-25
verified exercises	topic groups	source papers

English solutions aligned with the supplied official answers.

Each answer has two parts: the original Official Answer and the existing Supplementary Explanation. Source PDF pages are retained for verification.

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All verified, in-scope questions linked to Probability in the knowledge base.

Each question is included once, including cross-topic applications.

READING GUIDE

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How to use this booklet

Questions are grouped by their first Probability subtopic, then ordered by difficulty and year (newest first). Each question appears once as P001-P059. Difficulty ranges from 1 (introductory) to 4 (challenging).

Official Answer reproduces the answer in the source paper, with its physical PDF page reference.

Supplementary Explanation retains the existing independently authored reasoning, including the stated result and derivation. The two parts remain separate throughout the booklet.

Source references use physical PDF pages. File links work with the local source collection at its existing location.

The answer provenance shown for each exercise retains its knowledge-base classification; attaching an official quotation does not reclassify a derived correction as official.

P001

Introduction to Probability

2016 Final | Question 3(a) | Source page 7 | Difficulty 1
Q-2016-FINAL-P03-a-1 | Answer provenance: official

> A bag contains two dice. One dice is a normal six-sided fair dice, while the other is a special dice also with six sides where each of the 4 sides shows one dot and each of the remaining 2 sides shows two dots. Each side of both dice is equally likely to be on top when the corresponding dice is rolled. One of the dice is picked from the bag at random (i.e., the event that the fair dice is picked and the event that the special dice is picked are equally likely to occur), and this dice is rolled 2 times. Let A_1 be the event that the number of dots on top is smaller or equal to two dots on the first roll, and A_2 be the event that the number of dots on top is smaller than or equal to two dots on the second roll. We denote the complement of A_1 and the complement of A_2 by $\overline{A_1}$ and $\overline{A_2}$, respectively.
>
> (a) What is $P(A_1)$ and $P(A_2)$?
>
> Atomic focus [editorial]: $P(A_1)$; the second requested marginal is a separate atom.

Official Answer

Source: PDF physical page 7 (2016-final.pdf).

(a) If a fair dice is picked, the probability that we get one or two dots on top is $\frac{1}{3}$. If a special dice is picked, the probability that we get one or two dots on top is 1. Therefore, $P(A_1) = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot 1 = \frac{2}{3}$. By the same argument, $P(A_2) = \frac{2}{3}$.

Supplementary Explanation

Answer. $P(A_1) = 2/3$.

Derivation/proof. Let F and S denote selecting the fair and special dice. They partition the experiment and each has probability $1/2$. Total probability gives $P(A_1) = P(A_1 \mid F)P(F) + P(A_1 \mid S)P(S) = \frac{1}{3}\frac{1}{2} + 1\frac{1}{2} = \frac{2}{3}$.

The equal weights come from the random selection of the die, not from a uniform distribution over the resulting numerical values.

Explicitly, $P(A_1 \mid F) = 1/3$ and the weighted sum is $(1/2)(1/3) + (1/2)(1) = 2/3$.

P002

Introduction to Probability

2016 Final | Question 3(a) | Source page 7 | Difficulty 1
Q-2016-FINAL-P03-a-2 | Answer provenance: official

> A bag contains two dice. One dice is a normal six-sided fair dice, while the other is a special dice also with six sides where each of the 4 sides shows one dot and each of the remaining 2 sides shows two dots. Each side of both dice is equally likely to be on top when the corresponding dice is rolled. One of the dice is picked from the bag at random (i.e., the event that the fair dice is picked and the event that the special dice is picked are equally likely to occur), and this dice is rolled 2 times. Let A_1 be the event that the number of dots on top is smaller or equal to two dots on the first roll, and A_2 be the event that the number of dots on top is smaller than or equal to two dots on the second roll. We denote the complement of A_1 and the complement of A_2 by $\overline{A_1}$ and $\overline{A_2}$, respectively.
>
> (a) What is $P(A_1)$ and $P(A_2)$?
>
> Atomic focus [editorial]: $P(A_2)$; the first requested marginal is a separate atom.

Official Answer

Source: PDF physical page 7 (2016-final.pdf).

(a) If a fair dice is picked, the probability that we get one or two dots on top is $\frac{1}{3}$. If a special dice is picked, the probability that we get one or two dots on top is 1. Therefore, $P(A_1) = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot 1 = \frac{2}{3}$. By the same argument, $P(A_2) = \frac{2}{3}$.

Supplementary Explanation

Answer. $P(A_2) = 2/3$.

Derivation/proof. The choice of die is unchanged and unobserved. Conditional on the fair die, the second roll has probability $1/3$ of showing at most two; conditional on the special die that probability is 1. Thus $P(A_2) = \frac{1}{2}\frac{1}{3} + \frac{1}{2} = \frac{2}{3}$. Equality of the marginals does not establish independence of the events.

This question is not conditioned on the first roll. Thus $P(A_2) = (1/2)(1/3) + (1/2)(1) = 2/3$ still uses the original die weights.

P003

Introduction to Probability

2019 Final | Question 11(d) | Source page 13 | Difficulty 2
Q-2019-FINAL-P11-d-probability | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> What's its winning probability?

Official Answer

Source: physical PDF page 14 (2019-final.pdf).

We pick door 3 first, then always switch. The winning probability is $p(E) = p(W = 1, M = 2) + p(W = 2, M = 1) = 0.5 \cdot 1 + 0.3 \cdot 1 = 0.8$.

Supplementary Explanation

Answer. $0.8 = 4/5$ for initially choosing door 3 and always switching.

Derivation/proof. Under the proposed strategy, if $W = 1$, Monty opens door 2 and switching reaches door 1. If $W = 2$, he opens door 1 and switching reaches door 2. If $W = 3$, switching leaves the prize and loses regardless of his random choice. The three prize-location cases partition the experiment, so the win probability is $0.5 \cdot 1 + 0.3 \cdot 1 + 0.2 \cdot 0 = 0.8$.

When the prize is behind the initially selected third door, both possible host choices lead to a loss after switching, so the detailed randomization on that branch is irrelevant.

For the lowest-prior initial door, the win probability is $0.5 \cdot 1 + 0.3 \cdot 1 + 0.2 \cdot 0 = 0.8 = 4/5$. If the initial door actually contains the prize, both choices lose among the other two doors, explaining the final zero term.

P004

Introduction to Probability

2024 Spring Final | Question 5 | Source page 6 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P05 | Answer provenance: official

> You have a set of n biased coins. For each $m = 1, 2, \ldots, n$, the m-th coin has a probability of $\frac{1}{2m+1}$ of landing heads when tossed, and the outcomes of the tosses for each coin are independent of each other. Show that the probability of obtaining an odd number of heads after tossing all the n coins is equal to $\frac{n}{2n+1}$. Hint: One possible approach is to think in a recursive way.

Official Answer

Source: physical PDF page 6 (COMP2711_S2024_Final.pdf).

> Solution: Let P_n be the probability of obtaining an odd number of heads when tossing these n coins. In order to compute P_{n+1}, we need to consider two cases: Case 1: The $(n+1)$-th coin lands head and we have an even number of heads in the first n coins. Case 2: The $(n+1)$-th coin lands tail and we have an odd number of heads in the first n coins. Thus:
>
> $$P_{n+1} = \left(\frac{1}{2n+3}\right)(1 - P_n) + \left(1 - \frac{1}{2n+3}\right)P_n = \left(1 - \frac{2}{2n+3}\right)P_n + \frac{1}{2n+3}.$$
>
> Multiplying both sides by $(2n+3)$ we obtain $(2n+3)P_{n+1} = (2n+1)P_n + 1$. Substituting $A_n = (2n+1)P_n$, we can express this as $A_{n+1} = A_n + 1$. We know that $A_0 = 0$ because $P_0 = 0$. This implies that $A_n = n$, which in turn implies that $P_n = \frac{n}{2n+1}$, as desired.

Supplementary Explanation

Answer. $P_n = n/(2n+1)$.

Derivation/proof. Let P_n denote the odd-head probability for the first n coins, with $P_0 = 0$. Coin $n+1$ has head probability $1/(2n+3)$. An odd total comes from an even previous total and a head, or an odd previous total and a tail. These cases are mutually exclusive, and independence gives $P_{n+1} = \frac{1}{2n+3}(1 - P_n) + \frac{2n+2}{2n+3}P_n$. Hence $(2n+3)P_{n+1} = (2n+1)P_n + 1$. Set $A_n = (2n+1)P_n$. Then $A_{n+1} = A_n + 1$, $A_0 = 0$, so $A_n = n$ and $P_n = n/(2n+1)$. At $n = 1$ this gives $1/3$, as required.

P005

Introduction to Probability

2022 Final Part 2 | Question 6(c) | Source page 4 | Difficulty 3 | 4 marks
Q-2022-FINAL2-P06-c | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the probability that G with $n = 4$ vertices has 3 edges and is not connected.

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (c) The simple graph G with 4 vertices and 3 edges is not connected if and only if three edges form a triangle and the last vertex is disconnected, which happens with the probability $\binom{4}{3}p^3(1-p)^3$.

Supplementary Explanation

Answer. $4p^3(1-p)^3$.

Derivation/proof. A disconnected four-vertex simple graph with three edges cannot split2+2, since that allows at most two edges. Its nontrivial component must have three vertices, and all three possible edges must be present: a triangle. Choose the isolated vertex (or triangle vertices) in 4 ways. Each exact configuration has three present and three absent edges among K4's six possibilities, hence probability $p^3(1-p)^3$. These four configurations are disjoint, giving the result.

The structure is a triangle plus an isolated vertex among six potential edges. The four configurations are disjoint. This requires absence of all three edges to the fourth vertex, not merely the presence of a triangle.

P006

Introduction to Probability

2020 Final Part 2 | Question 5(d) | Source page 4 | Difficulty 3 | 3 marks
Q-2020-FINAL2-P05-d | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the probability that G with $n = 4$ vertices is connected and X takes its minimal possible value?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

G is connected and X is minimal if and only if G is a tree. The probability of having 3 edges in G is $\binom{6}{3}p^3(1-p)^3$. However, if the 3 edges form a triangle, then the last vertex will be disconnected, which happens with probability $\binom{4}{3}p^3(1-p)^3$. So, the final probability is

$$\binom{6}{3}p^3(1-p)^3 - \binom{4}{3}p^3(1-p)^3 = 16p^3(1-p)^3.$$

Supplementary Explanation

Answer. $16p^3(1-p)^3$.

Derivation/proof. A connected four-vertex graph has at least three edges, achieved exactly by trees. There are six potential edges and $\binom{6}{3} = 20$ three-edge subsets. A disconnected three-edge simple graph on four vertices must be a triangle plus an isolate: a 2+2 split permits at most two edges. There are $\binom{4}{3} = 4$ such triangles, leaving 16 trees. Each exact edge set has probability $p^3(1-p)^3$, so multiply by 16.

To verify the disconnected exception is exhaustive, a partition into two pairs permits at most two edges; finer partitions permit no more. Thus a disconnected three-edge graph must consist of a triangle and an isolated vertex.

Of the twenty three-edge subsets, exactly four form a triangle plus an isolated vertex. Removing them leaves $20 - 4 = 16$ trees, each with the same probability $p^3(1-p)^3$.

P007

Introduction to Probability

2025 Spring Final | Question 10 | Source page 13 | Difficulty 4 | 7 marks
Q-2025S-FINAL-P10 | Answer provenance: official

> A 3-by-3 unit-square grid is to be colored such that each unit square is independently colored red or blue with equal probability. What is the probability that the grid does not contain any 2-by-2 red square? Justify your answer.

Official Answer

Source: physical PDF page 13 (COMP2711_S2025_Final.pdf).

Since each of the 9 unit squares of the 3-by-3 square can take one of two colors, in total there are 2^9 different ways of coloring the 3-by-3 square.

Let us count now the colorings that form a 2-by-2 red square. This red square can be formed in the upper left, upper right, bottom left or bottom right corner of the 3-by-3 square. Let A, B, C, D be, respectively, the sets of the four types of colorings listed above that produce a 2-by-2 red square. Since the other 5 squares of the 3-by-3 square can have any color, we have $|A| = |B| = |C| = |D| = 2^5$. These sets are however not disjoint, so we must apply inclusion-exclusion:

$$\begin{aligned}|A \cup B \cup C \cup D| = {} & |A| + |B| + |C| + |D| \\ & - |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| \\ & + |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \\ & - |A \cap B \cap C \cap D|.\end{aligned}$$

As $A \cap B$ represents the set of the colorings having the first two rows red (and the 3 squares of the lowest row free to take any of the two colors), we have $|A \cap B| = 2^3$. Similarly, $|A \cap C| = 2^3$, $|B \cap D| = 2^3$ and $|C \cap D| = 2^3$. We have a different situation for $A \cap D$ and $B \cap C$: if two opposing 2-by-2 squares are red, this leaves uncertain only the colors of the two remaining 1-by-1 corners, therefore $|A \cap D| = |B \cap C| = 2^2$. The intersections of 3 of the sets leaves undecided only the color of one square, therefore each of these intersections has 2 elements. Finally, the intersection of all the four sets contains only the square completely colored in red, that is, one element. We obtain:

$$|A \cup B \cup C \cup D| = 4 \cdot 2^5 - (4 \cdot 2^3 + 2 \cdot 2^2) + 4 \cdot 2 - 1 = 95.$$

So from the 512 possible colorings, $512 - 95 = 417$ do not contain any 2-by-2 red square, so the probability of not having a 2-by-2 red square is $\frac{417}{512}$.

Supplementary Explanation

Answer. $417/512$.

Derivation/proof. The nine independent fair colors give $2^9 = 512$ equally likely outcomes. Events A,B,C,D require the top-left, top-right, bottom-left and bottom-right 2×2 squares to be all red. Each fixes four cells and permits $2^5 = 32$ colorings. Of the six pair intersections, four adjacent pairs fix six cells and permit eight colorings each; two diagonal pairs fix seven and permit four each. Any three events fix eight cells, yielding two colorings for each of four triple intersections. All four events require the unique all-red grid. Inclusion-exclusion gives $4 \cdot 32 - (4 \cdot 8 + 2 \cdot 4) + 4 \cdot 2 - 1 = 95$ colorings with a red square. The complement has $512 - 95 = 417$, giving probability $417/512$.

P008

Conditional Probability

2022 Final Part 2 | Question 2(a) | Source page 2 | Difficulty 1
Q-2022-FINAL2-P02-a-23 | Answer provenance: official

> Consider the following 5-door version of the Monty Hall problem. There are 5 doors, behind one of which (randomly with equal probability) there is a car (which you want), and behind the rest of which there are goats (which you don't want). You choose a door. Monty then opens 2 goat doors. Assume that Monty knows which door has the car, and will choose with equal probabilities from all his choices of which goat doors to open. Without loss of generality, assume your first guess is door 1. Let W be the random variable denoting the door with the car and M the random variable denoting the two doors that Monty opens. Note that M is a random variable whose value is a subset of $\{1, 2, 3, 4, 5\}$.
>
> What is $P(M = \{2, 3\} \mid W = 1)$?

Official Answer

Source: physical PDF page 2 (2022-final-part-2.pdf).

> (a) $P(M = \{2, 3\} \mid W = 1) = P(M = \{3, 5\} \mid W = 1) = 1/\binom{4}{2} = 1/6$.

Supplementary Explanation

Answer. $1/6$.

Derivation/proof. Given W=1, all four other doors 2,3,4,5 contain goats and remain eligible for Monty's opening. He selects an unordered two-door subset uniformly; there are C(4,2)=6 such pairs. {2,3} is one pair, so its conditional probability is 1/6. The two openings are not ordered outcomes.

There are $\binom{4}{2} = 6$ pairs and $P(M = \{2, 3\} \mid W = 1) = 1/6$. The eligible goat-door set is the same under either interpretation of whether the initially chosen door may be opened, because that door contains the car.

P009

Conditional Probability

2022 Final Part 2 | Question 2(a) | Source page 2 | Difficulty 1
Q-2022-FINAL2-P02-a-35 | Answer provenance: official

> Consider the following 5-door version of the Monty Hall problem. There are 5 doors, behind one of which (randomly with equal probability) there is a car (which you want), and behind the rest of which there are goats (which you don't want). You choose a door. Monty then opens 2 goat doors. Assume that Monty knows which door has the car, and will choose with equal probabilities from all his choices of which goat doors to open. Without loss of generality, assume your first guess is door 1. Let W be the random variable denoting the door with the car and M the random variable denoting the two doors that Monty opens. Note that M is a random variable whose value is a subset of $\{1, 2, 3, 4, 5\}$.
>
> What is $P(M = \{3, 5\} \mid W = 1)$?

Official Answer

Source: physical PDF page 2 (2022-final-part-2.pdf).

> (a) $P(M = \{2, 3\} \mid W = 1) = P(M = \{3, 5\} \mid W = 1) = 1/\binom{4}{2} = 1/6.$

Supplementary Explanation

Answer. $1/6$.

Derivation/proof. With the car fixed behind door 1, Monty can choose any pair from goat doors 2,3,4,5. Uniformity assigns equal weight1/C(4,2) to each of the six pairs, including {3,5}. This is a fresh pair selection event under the same condition, not a sequential probability after already opening {2,3}.

The pairs are $\{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\}$. The target occupies one of six pairs. Both interpretations of the initially chosen door give the same set when the car is at one.

P010

Conditional Probability

2022 Final Part 2 | Question 2(b) | Source page 2 | Difficulty 1 | 2 marks
Q-2022-FINAL2-P02-b | Answer provenance: official

> Consider the following 5-door version of the Monty Hall problem. There are 5 doors, behind one of which (randomly with equal probability) there is a car (which you want), and behind the rest of which there are goats (which you don't want). You choose a door. Monty then opens 2 goat doors. Assume that Monty knows which door has the car, and will choose with equal probabilities from all his choices of which goat doors to open. Without loss of generality, assume your first guess is door 1. Let W be the random variable denoting the door with the car and M the random variable denoting the two doors that Monty opens. Note that M is a random variable whose value is a subset of $\{1, 2, 3, 4, 5\}$.
>
> What is $P(M = \{2, 3\} \mid W = 2)$?

Official Answer

Source: physical PDF page 2 (2022-final-part-2.pdf).

> (b) $P(M = \{2, 3\} \mid W = 2) = 0$

Supplementary Explanation

Answer. 0.

Derivation/proof. The condition W=2 places the car behind door 2. The event M={2,3} requires Monty to open that car door, contrary to the stated rule that he opens goat doors. Hence no permitted choice realizes this event and its conditional probability is zero, regardless of the number of other eligible pairs.

Thus $P(M = \{2, 3\} \mid W = 2) = 0$, independently of whether the initially chosen goat door may be opened. No count of the other eligible pairs is needed.

P011

Conditional Probability

2025 Fall Final | Question 3(a) | Source page 7 | Difficulty 2 | 3 marks
Q-2025F-FINAL-P03-a | Answer provenance: official

> We have 14 volleyball players, all of different heights. All 14! permutations of rankings (tallest = rank 1 to shortest = rank 14) are equally likely. Answer the following questions and briefly explain. You do not need to calculate the final numerical answer.
>
> (a) [3 points] Given A is taller than B, what is the probability that A is tallest?

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Final.pdf).

(a) Given $A > B$, permutations satisfying this: $14!/2$. If A is tallest, arrange others: 13! permutations.

$$\Pr(A \text{ tallest} \mid A > B) = \frac{13!}{14!/2} = \frac{2}{14} = \frac{1}{7}.$$

Supplementary Explanation

Answer. $1/7$.

Derivation/proof. All 14! rankings are equally likely. Swapping A and B gives a bijection between rankings with $A > B$ and those with $B > A$, so the conditional sample space contains $14!/2$ rankings. If A is first, $A > B$ holds automatically and the other 13 players have 13! arrangements. Thus the conditional probability is $13!/(14!/2) = 2/14 = 1/7$. The denominator counts rankings satisfying the condition, not all rankings.

P012

Conditional Probability

2023 Fall Midterm | Question 8 | Source page 17 | Difficulty 2 | 8 marks
Q-2023F-MID-P08 | Answer provenance: official

> Two computers, a sender and a receiver, are communicating with each other. The sender transmits bit 1 with a probability of 30% and bit 0 with a probability of 70%. The communication channel is noisy, and we know the following information about the transition of data:
>
> 1. If the sent bit is 0, there is a 90% probability that the receiver will receive bit 0 and a 10% probability that it will receive bit 1.
> 2. If the sent bit is 1, there is an 80% probability that the receiver will receive bit 1 and a 20% probability that it will receive bit 0.
>
> Given that the receiver computer received bit 0, determine the probability that the bit being sent was also 0.

Official Answer

Source: physical PDF page 17 (COMP2711_F2023_Midterm.pdf).

> Denote the bit sent by the sender to be S and the bit received by the receiver to be R. We know that:
>
> $$p(S=1)=0.3,\ p(S=0)=0.7$$
> $$p(R=1\mid S=1)=0.8,\ p(R=0\mid S=1)=0.2$$
> $$p(R=1\mid S=0)=0.1,\ p(R=0\mid S=0)=0.9$$
>
> By the Bayes' theorem,
>
> $$\begin{aligned} p(S=0\mid R=0) &= \frac{p(R=0\mid S=0)p(S=0)}{p(R=0)} \\ &= \frac{p(R=0\mid S=0)p(S=0)}{p(R=0\mid S=1)p(S=1)+p(R=0\mid S=0)p(S=0)} \\ &= \frac{0.9\times 0.7}{0.2\times 0.3+0.9\times 0.7} \\ &= \frac{21}{23} \\ &= 0.913. \end{aligned}$$

Supplementary Explanation

Answer. $P(S=0\mid R=0)=\dfrac{21}{23}\approx 0.91304$.

Derivation/proof. Let S be the sent bit and R the received bit. The joint probability of a sent zero and received zero is $P(S=0,R=0)=0.7\cdot 0.9=0.63$. Total probability gives $P(R=0)=0.7\cdot 0.9+0.3\cdot 0.2=0.69$. Dividing the joint event by the conditioning event yields $P(S=0\mid R=0)=0.63/0.69=21/23$. The alternative posterior is $0.06/0.69=2/23$, so the two possible sent-bit posteriors sum to 1.

The second joint probability is $0.3\cdot 0.2=0.06$. The source's final decimal is rounded; the exact value is $21/23$.

P013

Conditional Probability

2020 Final Part 2 | Question 3 | Source page 2 | Difficulty 2 | 5 marks
Q-2020-FINAL2-P03 | Answer provenance: official

> Let A and B be two events. It is known that $p(A) = 0.7$, $p(B) = 0.6$, $p(A \cap B) = 0.5$, what is $p(B \mid A \cup \overline{B})$?

Official Answer

Source: physical PDF page 2 (2020-final-part-2.pdf).

$$p(B \mid A \cup \overline{B}) = \frac{p(B \cap (A \cup \overline{B}))}{p(A \cup \overline{B})}.$$

Since $B \cap (A \cup \overline{B}) = (B \cap A) \cup (B \cap \overline{B}) = (B \cap A) \cup \varnothing = B \cap A$,

$$p(B \cap (A \cup \overline{B})) = p(B \cap A) = 0.5.$$

$$p(A \cup \overline{B}) = 1 - p(\overline{A} \cap B) = 1 - (p(B) - p(A \cap B)) = 0.9.$$

The final result is,

$$p(B \mid A \cup \overline{B}) = \frac{p(B \cap (A \cup \overline{B}))}{p(A \cup \overline{B})} = \frac{5}{9}.$$

Supplementary Explanation

Answer. $5/9$.

Derivation/proof. The conditional numerator is $p(B \cap (A \cup \overline{B})) = p(A \cap B) = .5$, since $B \cap \overline{B} = \varnothing$. The conditioning event excludes exactly $\overline{A} \cap B$, of probability $.6 - .5 = .1$. Its probability is therefore $.9$. Dividing gives $.5/.9 = 5/9$. The bar over B is visible in the printed prompt and materially changes both event calculations.

The probability of the excluded event is $0.6 - 0.5 = 0.1$, leaving conditioning probability 0.9. The conditional numerator is 0.5, hence the quotient is $0.5/0.9 = 5/9$.

P014

Conditional Probability

2020 Final Part 2 | Question 5(a) | Source page 3 | Difficulty 2 | 2 marks
Q-2020-FINAL2-P05-a | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> Suppose we remove a vertex u and all edges incident to u, and get a subgraph G'. Given G' is connected, what is the probability that G is also connected?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

G is also connected if and only if u connects with at least one vertex in G'. Since we know u links to any other vertex with probability p, the probability that it has no edge with other vertices is, $(1-p)^{n-1}$. So the probability that G is connected is $1-(1-p)^{n-1}$.

Supplementary Explanation

Answer. $1-(1-p)^{n-1}$ whenever the conditioning event has positive probability.

Derivation/proof. With G′ connected, G is connected exactly when u has at least one neighbor among the n−1 remaining vertices. The event about G′ depends only on edges not incident to u, so conditioning does not change the independent incident edge probabilities. All n−1 incident edges are absent with probability (1−p)^(n−1). Taking its complement yields the answer.

As usual, the conditioning event must have positive probability; the formula does not define conditioning on a zero-probability event.

P015

Conditional Probability

2019 Final | Question 11(a) | Source page 13 | Difficulty 2
Q-2019-FINAL-P11-a-decision | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> Based on your calculation, should you switch to door 3?

Official Answer

Source: physical PDF page 13 (2019-final.pdf); physical PDF page 14 (2019-final.pdf).

By the Bayes' theorem,

$$p(W = w \mid M = 2) = \frac{p(M = 2 \mid W = w)p(W = w)}{p(M = 2)},$$

where $p(M = 2) = \sum_{w=1}^{3} p(M = 2 \mid W = w)p(W = w)$. We have $p(W = 1) = 0.5, p(W = 2) = 0.3, p(W = 3) = 0.2, p(M = 2 \mid W = 1) = 1/2, p(M = 2 \mid W = 2) = 0, p(M = 2 \mid W = 3) = 1$. Plugging in these numbers, we obtain $p(M = 2) = 0.45$ and $p(W = 1 \mid M = 2) = 5/9, p(W = 3 \mid M = 2) = 4/9$. So, we should not switch. Contrary to our intuition, the probability that our initial guess is correct has increased after door 2 is opened!

Supplementary Explanation

Answer. No; keep door 1.

Derivation/proof. The joint masses compatible with opening door 2 are 0.25 for prize door 1 and 0.2 for prize door 3, summing to 0.45. Their posterior probabilities are $5/9$ and $4/9$. Keeping wins with $5/9$, while switching wins with $4/9$, so keeping is strictly better under this observation.

The two posterior winning probabilities satisfy $5/9 > 4/9$, so keep door one after this observation.

P016

Conditional Probability

2019 Final | Question 11(a) | Source page 13 | Difficulty 2
Q-2019-FINAL-P11-a-W1 | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$

Official Answer

Source: physical PDF page 13 (2019-final.pdf); physical PDF page 14 (2019-final.pdf).

By the Bayes' theorem,

$$p(W = w \mid M = 2) = \frac{p(M = 2 \mid W = w)p(W = w)}{p(M = 2)},$$

where $p(M = 2) = \sum_{w=1}^{3} p(M = 2 \mid W = w)p(W = w)$. We have $p(W = 1) = 0.5, p(W = 2) = 0.3, p(W = 3) = 0.2, p(M = 2 \mid W = 1) = 1/2, p(M = 2 \mid W = 2) = 0, p(M = 2 \mid W = 3) = 1$. Plugging in these numbers, we obtain $p(M = 2) = 0.45$ and $p(W = 1 \mid M = 2) = 5/9, p(W = 3 \mid M = 2) = 4/9$. So, we should not switch. Contrary to our intuition, the probability that our initial guess is correct has increased after door 2 is opened!

Supplementary Explanation

Answer. $5/9$.

Derivation/proof. Monty opens door 2 with probability $1/2$ if $W = 1$, zero if $W = 2$, and one if $W = 3$. Therefore $P(M = 2) = 0.5(1/2) + 0.3(0) + 0.2(1) = 0.45$. The joint mass for $W = 1, M = 2$ is 0.25. Bayes gives $0.25/0.45 = 5/9$.

P017

Conditional Probability

2019 Final | Question 11(a) | Source page 13 | Difficulty 2
Q-2019-FINAL-P11-a-W3 | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> and $p(W = 3 \mid M = 2)$?

Official Answer

Source: physical PDF page 13 (2019-final.pdf); physical PDF page 14 (2019-final.pdf).

By the Bayes' theorem,

$$p(W = w \mid M = 2) = \frac{p(M = 2 \mid W = w)p(W = w)}{p(M = 2)},$$

where $p(M = 2) = \sum_{w=1}^{3} p(M = 2 \mid W = w)p(W = w)$. We have $p(W = 1) = 0.5, p(W = 2) = 0.3, p(W = 3) = 0.2, p(M = 2 \mid W = 1) = 1/2, p(M = 2 \mid W = 2) = 0, p(M = 2 \mid W = 3) = 1$. Plugging in these numbers, we obtain $p(M = 2) = 0.45$ and $p(W = 1 \mid M = 2) = 5/9, p(W = 3 \mid M = 2) = 4/9$. So, we should not switch. Contrary to our intuition, the probability that our initial guess is correct has increased after door 2 is opened!

Supplementary Explanation

Answer. $4/9$.

Derivation/proof. If the prize is at door 3, Monty must open door 2, so $P(W = 3, M = 2) = 0.2$. Total probability gives $P(M = 2) = 0.5(1/2) + 0.3(0) + 0.2(1) = 0.45$. Thus $P(W = 3 \mid M = 2) = 0.2/0.45 = 4/9$. The other closed-door posterior is $5/9$, and they sum to one.

The two posterior probabilities sum to $5/9 + 4/9 = 1$; the open door has probability zero.

Door two is opened with total probability $0.2 + 0.5/2 = 0.45$: the prize-at-door-three branch forces that opening, while the prize-at-door-one branch chooses it with probability one half.

P018

Conditional Probability

2019 Final | Question 11(b) | Source page 13 | Difficulty 2
Q-2019-FINAL-P11-b-decision | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> In this case, should you switch to door 2?

Official Answer

Source: physical PDF page 13 (2019-final.pdf); physical PDF page 14 (2019-final.pdf).

Same as above, but with $p(M = 3 \mid W = 1) = 1/2, p(M = 3 \mid W = 2) = 1, p(M = 3 \mid W = 3) = 0$. Plugging in these numbers, we obtain $p(M = 3) = 0.55$ and $p(W = 1 \mid M = 3) = 5/11, p(W = 2 \mid M = 3) = 6/11$. So, we should switch to door 2. This time, the probability that our initial guess is correct has decreased.

Supplementary Explanation

Answer. Yes; switch to door 2.

Derivation/proof. For the observation $M = 3$, the posterior probabilities are $P(W = 1 \mid M = 3) = 0.25/0.55 = 5/11$ and $P(W = 2 \mid M = 3) = 0.3/0.55 = 6/11$. Switching therefore raises the conditional win probability from $5/11$ to $6/11$. The optimal response differs from the response when door 2 is opened.

Keeping the initial door wins with probability $(0.5/2)/0.55 = 5/11$. Since $6/11 > 5/11$, switching to door two is better after this opening.

P019

Conditional Probability

2019 Final | Question 11(b) | Source page 13 | Difficulty 2
Q-2019-FINAL-P11-b-W1 | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$

Official Answer

Source: physical PDF page 13 (2019-final.pdf); physical PDF page 14 (2019-final.pdf).

Same as above, but with $p(M = 3 \mid W = 1) = 1/2, p(M = 3 \mid W = 2) = 1, p(M = 3 \mid W = 3) = 0$. Plugging in these numbers, we obtain $p(M = 3) = 0.55$ and $p(W = 1 \mid M = 3) = 5/11, p(W = 2 \mid M = 3) = 6/11$. So, we should switch to door 2. This time, the probability that our initial guess is correct has decreased.

Supplementary Explanation

Answer. $5/11$.

Derivation/proof. Here $P(M = 3 \mid W = 1) = 1/2$, $P(M = 3 \mid W = 2) = 1$, and $P(M = 3 \mid W = 3) = 0$. Hence $P(M = 3) = 0.5(1/2) + 0.3 = 0.55$. The joint mass $P(W = 1, M = 3) = 0.25$, giving posterior $0.25/0.55 = 5/11$.

The total probability of the observed opening is $0.5(1/2) + 0.3(1) + 0.2(0) = 0.55$. These terms condition on the three mutually exclusive prize locations.

P020

Conditional Probability

2019 Final | Question 11(b) | Source page 13 | Difficulty 2
Q-2019-FINAL-P11-b-W2 | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> and $p(W = 2 \mid M = 3)$?

Official Answer

Source: physical PDF page 13 (2019-final.pdf); physical PDF page 14 (2019-final.pdf).

Same as above, but with $p(M = 3 \mid W = 1) = 1/2, p(M = 3 \mid W = 2) = 1, p(M = 3 \mid W = 3) = 0$. Plugging in these numbers, we obtain $p(M = 3) = 0.55$ and $p(W = 1 \mid M = 3) = 5/11, p(W = 2 \mid M = 3) = 6/11$. So, we should switch to door 2. This time, the probability that our initial guess is correct has decreased.

Supplementary Explanation

Answer. $6/11$.

Derivation/proof. When the prize is behind door 2 and you chose door 1, Monty must open door 3, giving joint probability 0.3. The total opening probability is $0.5(1/2) + 0.3(1) + 0.2(0) = 0.55$. Therefore the posterior is $0.3/0.55 = 6/11$, complementary to the $5/11$ for door 1.

The joint event is $P(W = 2, M = 3) = 0.3$, and its observation probability is $0.3 + 0.25 = 0.55$. Dividing the joint probability by this total gives the posterior for door two.

P021

Conditional Probability

2017 Final | Question 5(a) | Source page 8 | Difficulty 2 | 4 marks
Q-2017-FINAL-P05-a | Answer provenance: official

> You play against your friend in a coin flipping game, where the objective is to get the most heads after three coin flips. A player wins if they have more heads than the opponent. If the numbers of heads are equal, then no one wins; it is a tie. You will take turns flipping coins, and your friend flips first.
>
> You want to cheat, so you created a fake coin that looks identical to a real, fair coin. The fake coin has a 80% chance of getting heads. Your friend is suspicious, so your friend gets to pick first from the two coins, randomly, with equal probability. You are forced to take the other coin. You will both use your own coin for all three coin flips. The game begins, and your friend flips the coin once and gets heads.
>
> (a) Given that your friend's first coin flip returns heads, what is the probability that your friend got the fake coin, to your disadvantage?

Official Answer

Source: physical PDF page 8 (2017-final.pdf); physical PDF page 9 (2017-final.pdf).

Define the following events: E_1: Your friend got the fake coin. E_2: Your friend got the real coin. H: Your friend's coin flip returns heads. T: Your friend's coin flip returns tails. We want to know $p(E_1 \mid H)$. We know that $p(H) + p(T) = 1, p(E_1) + p(E_2) = 1$. Then we have:

$$p(E_1 \mid H) \cdot p(H) = p(H \mid E_1) \cdot p(E_1)$$

$$p(E_1 \mid H) = 0.8 \cdot 0.5 / p(H) = 0.4 / (0.5 \cdot 0.8 + 0.5 \cdot 0.5) = 0.4 / 0.65 = 8/13.$$

Supplementary Explanation

Answer. $8/13$.

Derivation/proof. Let F mean the friend holds the fake coin and H mean the observed head. Then $P(F) = 1/2$, $P(H \mid F) = 4/5$, and $P(H \mid F^c) = 1/2$. Total probability gives $P(H) = (1/2)(4/5) + (1/2)(1/2) = 13/20$. Bayes gives $P(F \mid H) = (2/5)/(13/20) = 8/13$.

The joint probability is $P(F \cap H) = 2/5$. Only the observed first toss enters this calculation; no assumption about later tosses is needed.

P022

Conditional Probability

2016 Final | Question 3(e) | Source page 7 | Difficulty 2 | 3 marks
Q-2016-FINAL-P03-e | Answer provenance: derived

> A bag contains two dice. One dice is a normal six-sided fair dice, while the other is a special dice also with six sides where each of the 4 sides shows one dot and each of the remaining 2 sides shows two dots. Each side of both dice is equally likely to be on top when the corresponding dice is rolled. One of the dice is picked from the bag at random (i.e., the event that the fair dice is picked and the event that the special dice is picked are equally likely to occur), and this dice is rolled 2 times. Let A_1 be the event that the number of dots on top is smaller or equal to two dots on the first roll, and A_2 be the event that the number of dots on top is smaller than or equal to two dots on the second roll. We denote the complement of A_1 and the complement of A_2 by $\overline{A_1}$ and $\overline{A_2}$, respectively.
>
> (e) Are A_1 and A_2 independent? Explain your answer.

Official Answer

Source: PDF physical page 7 (2016-final.pdf).

(e) No, A_1 and A_2 are dependent because $P(A_2 \mid A_1) \neq P(A_2)$.

Supplementary Explanation

Answer. No; A_1 and A_2 are dependent for every joint law compatible with the printed conditions.

Derivation/proof. Let F denote selection of the fair die and set $t = P(A_1 \cap A_2 \mid F)$. Each fair-die marginal is $1/3$, so $0 \leq t \leq 1/3$. On the special die both events always occur. With the given equal die priors, $P(A_i) = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} = 2/3$ for each i. But $P(A_1 \cap A_2) = \frac{1}{2}t + \frac{1}{2} \geq 1/2 > 4/9 = P(A_1)P(A_2)$, which rules out independence. No exact conditional intersection value or independence of repeated rolls was assumed.

The bound $t \leq 1/3$ follows because the intersection is contained in either fair-die event. The proof never fixes $t = 1/9$ and does not borrow the quarantined exact values $5/6$ or $5/9$.

The general-coupling joint probability is $\frac{1}{2}t + \frac{1}{2}$, which is at least one-half for every allowed t.

P023

Conditional Probability

2025 Fall Final | Question 3(b) | Source page 7 | Difficulty 3 | 3 marks
Q-2025F-FINAL-P03-b | Answer provenance: official

> We have 14 volleyball players, all of different heights. All 14! permutations of rankings (tallest = rank 1 to shortest = rank 14) are equally likely. Answer the following questions and briefly explain. You do not need to calculate the final numerical answer.
>
> (b) [3 points] Given A is taller than B, what is the probability that A is the third tallest?

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Final.pdf).

(b) Given $A > B$. If A is rank 3, B must be in ranks $4, \ldots, 14$: 11 choices. Then arrange remaining 12 players: 12!.

$$\Pr(A \text{ rank } 3 \mid A > B) = \frac{11 \cdot 12!}{14!/2} = \frac{11 \cdot 2}{14 \cdot 13} = \frac{11}{91}.$$

Supplementary Explanation

Answer. $11/91$.

Derivation/proof. The condition $A > B$ leaves $14!/2$ equally likely rankings. Fix A in third place. Then B can occupy ranks 4 through 14, giving $14 - 3 = 11$ choices. After fixing both positions, the remaining 12 players have 12! arrangements, so there are $11 \cdot 12!$ favorable rankings. Division gives $11 \cdot 12!/(14!/2) = 22/(14 \cdot 13) = 11/91$. Neither position above A may be given to B.

P024

Conditional Probability

2025 Fall Final | Question 3(c) | Source page 7 | Difficulty 3 | 3 marks
Q-2025F-FINAL-P03-c | Answer provenance: official

> We have 14 volleyball players, all of different heights. All 14! permutations of rankings (tallest = rank 1 to shortest = rank 14) are equally likely. Answer the following questions and briefly explain. You do not need to calculate the final numerical answer.
>
> (c) [3 points] Given A is taller than B and shorter than C, what is the probability that A is seventh tallest?

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Final.pdf).

(c) Given $C > A > B$. Permutations satisfying this: $14!/6$. If A is rank 7, C in ranks $1, \ldots, 6$ (6 choices), B in ranks $8, \ldots, 14$ (7 choices), arrange remaining 11: 11!.

$$\Pr(A \text{ rank } 7 \mid C > A > B) = \frac{6 \cdot 7 \cdot 11!}{14!/6} = \frac{252}{14 \cdot 13 \cdot 12} = \frac{3}{26}.$$

Supplementary Explanation

Answer. $3/26$.

Derivation/proof. The three distinct-height players have $3! = 6$ equally likely relative orders, so $C > A > B$ leaves $14!/6$ rankings. Fix A seventh. Player C has six choices among ranks 1–6; player B has seven choices among ranks 8–14. The other 11 players can be arbitrarily arranged, giving $6 \cdot 7 \cdot 11!$ favorable rankings. Hence the conditional probability is $6 \cdot 7 \cdot 11!/(14!/6) = 252/(14 \cdot 13 \cdot 12) = 3/26$. The position sets for C and B are disjoint, so multiplication does not create collisions.

P025

Conditional Probability

2025 Spring Final | Question 11 | Source page 15 | Difficulty 3 | 7 marks
Q-2025S-FINAL-P11 | Answer provenance: official

> A man is known to tell the truth with probability $\frac{3}{4}$ (independent of the event outcome) and can only respond with “YES” or “NO”. He rolls a fair six-sided die and is then asked whether the die landed on 6. He responds, “YES”! What is the probability that the die actually landed on 6? Justify your answer.

Official Answer

Source: physical PDF page 15 (COMP2711_S2025_Final.pdf).

There are two scenarios in which the man responds YES: either he is being truthful, or he is not. Note that his truthfulness is independent of the die's outcome.

Let S be the event of rolling a six, $P(S) = \frac{1}{6}$.

Let $\overline{S}$ be the event of not rolling a six, $P(\overline{S}) = \frac{5}{6}$.

Let T be the event that the man tells the truth, regardless of the outcome, $P(T) = \frac{3}{4}$.

Let $\overline{T}$ be the event that the man does not tell the truth, regardless of the outcome, $P(\overline{T}) = \frac{1}{4}$.

Let M be the event that the man responds "YES" when asked if the die landed on 6. We want to find $P(S \mid M)$.

First Solution: Using the Bayes' theorem:

$$P(S \mid M) = \frac{P(M \mid S)P(S)}{P(M \mid S)P(S) + P(M \mid \overline{S})P(\overline{S})} = \frac{\frac{3}{4} \cdot \frac{1}{6}}{\frac{3}{4} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{5}{6}} = \frac{3}{8}.$$

Second Solution:

$$P(M) = P(S \cap T) + P(\overline{S} \cap \overline{T}) = \frac{1}{6} \cdot \frac{3}{4} + \frac{5}{6} \cdot \frac{1}{4} = \frac{8}{24}$$

$$P(S \cap M) = \frac{3}{4} \cdot \frac{1}{6} = \frac{3}{24}$$

$$P(S \mid M) = \frac{P(S \cap M)}{P(M)} = \frac{3/24}{8/24} = \frac{3}{8}$$

Supplementary Explanation

Answer. $3/8$.

Derivation/proof. Let S mean a six and M mean YES. Truthfulness is explicitly independent of the outcome, so $P(S) = 1/6$, $P(M \mid S) = 3/4$, and $P(M \mid \overline{S}) = 1/4$. The disjoint, exhaustive sources of YES give $P(M) = \frac{1}{6}\frac{3}{4} + \frac{5}{6}\frac{1}{4} = 8/24$. Rolling a six and saying YES has probability $P(S \cap M) = 3/24$. Thus $P(S \mid M) = (3/24)/(8/24) = 3/8$. This is lower than the truthfulness rate because the prior probability of a non-six is much larger.

P026

Conditional Probability

2023 Fall Final | Question 3 | Source page 6 | Difficulty 3 | 7 marks
Q-2023F-FINAL-P03 | Answer provenance: official

> Suppose that one person in 10,000 people has a rare genetic disease. There is an excellent test for the disease; 99.9% of people with the disease test positive and only 0.02% who do not have the disease test positive. What is the probability that someone who tests negative does not have the disease? You have to show all the steps. But you do not have to calculate the final numerical values in each step.

Official Answer

Source: physical PDF page 6 (COMP2711_F2023_Final.pdf).

> First, we determine these events: A is the event that someone has the disease, and B is the event that someone's test is positive. From the problem we know:
>
> $$P(A) = \frac{1}{10000}, \qquad P(B \mid A) = 0.999, \qquad P(B \mid\sim A) = 0.0002.$$
>
> $$P(B) = P(A) * P(B \mid A) + P(\sim A) * P(B \mid\sim A) = \frac{1}{10000} \cdot 0.999 + (1 - \frac{1}{10000}) * 0.0002 =$$
>
> What is the probability that someone who tests negative does not have the disease? So we want to calculate: $P(\sim A \mid\sim B) = \frac{P(\sim B \mid\sim A)P(\sim A)}{P(\sim B)}$ in which $P(\sim B \mid\sim A) = 1 - P(B \mid\sim A)$ and $P(\sim B) = 1 - P(B)$ and $P(\sim A) = 1 - P(A)$.

Supplementary Explanation

Answer. $\dfrac{(1 - 1/10000)(1 - 0.0002)}{(1 - 1/10000)(1 - 0.0002) + (1/10000)(1 - 0.999)} = \dfrac{49\,985\,001}{49\,985\,006}.$

Derivation/proof. Let D mean disease and N mean a negative test. The given probabilities imply $P(D) = 0.0001$, $P(N \mid D) = 0.001$, and $P(N \mid D^c) = 0.9998$. Thus $P(N \cap D^c) = 0.9999 \cdot 0.9998 = 0.99970002$, while $P(N \cap D) = 0.0001 \cdot 0.001 = 0.0000001$. Total probability gives $P(N) = 0.99970012$. Bayes then yields $P(D^c \mid N) = 0.99970002/0.99970012 = 49\,985\,001/49\,985\,006$, a value slightly below 1.

The negative-result probability is 0.99970012, and the posterior is $49\,985\,001/49\,985\,006$.

P027

Conditional Probability

2019 Final | Question 11(c) | Source page 13 | Difficulty 3 | 2 marks
Q-2019-FINAL-P11-c | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> (c) What's the winning probability of your strategy above?

Official Answer

Source: physical PDF page 13 (2019-final.pdf); physical PDF page 14 (2019-final.pdf).

Let E denote the event that you win. We have $p(E) = \sum_{m=2}^{3} p(M = m)p(E \mid M = m) = 0.45 \cdot 5/9 + 0.55 \cdot 6/11 = 0.55$. Alternatively, $p(E) = p(W = 1, M = 2) + p(W = 2, M = 3) = 0.5 \cdot 0.5 + 0.3 \cdot 1 = 0.55$.

Supplementary Explanation

Answer. $0.55 = 11/20$.

Derivation/proof. Keep door 1 when $M = 2$, and choose door 2 when $M = 3$. These disjoint winning events are $(W = 1, M = 2)$ with mass $0.5(1/2) = 0.25$, and $(W = 2, M = 3)$ with mass $0.3(1) = 0.3$. Thus the overall win probability is $0.25 + 0.3 = 0.55$. Equivalently total probability gives $0.45(5/9) + 0.55(6/11) = 0.55$.

The complete strategy wins in branches of total probability $0.25 + 0.3 = 0.55 = 11/20$. Equivalently, average the conditional optimal outcomes: $0.45 \cdot (5/9) + 0.55 \cdot (6/11) = 0.55$.

P028

Conditional Probability

2019 Final | Question 11(d) | Source page 13 | Difficulty 3
Q-2019-FINAL-P11-d-strategy | Answer provenance: official

> Recall that we gave two explanations for the Monty Hall game. The first is an intuitive one: Our initial guess is correct with probability $\frac{1}{3}$, which is not affected by Monty opening another door, so switching will give us a winning probability of $\frac{2}{3}$. Then we gave a formal proof that $p(W = 1 \mid M = 2) = p(W = 1) = \frac{1}{3}$, where W denotes the winning door and M the door Monty opens. Below, we will see that the intuitive explanation only "happens" to be correct in this case, and can easily go wrong if we slightly perturb the probabilities.
>
> Now, suppose that the probabilities of the 3 doors having the prize are 0.5, 0.3, and 0.2, respectively. Then obviously we should pick door 1 as our initial guess. Then Monty will open one of door 2 or 3 without the prize. If neither has the prize, he will pick one of them randomly with equal probability.
>
> (a) Suppose $M = 2$, i.e., Monty opens door 2. What is $p(W = 1 \mid M = 2)$ and $p(W = 3 \mid M = 2)$? Based on your calculation, should you switch to door 3?
>
> (b) Now suppose $M = 3$, i.e., Monty opens door 3. What is $p(W = 1 \mid M = 3)$ and $p(W = 2 \mid M = 3)$? In this case, should you switch to door 2?
>
> (c) What's the winning probability of your strategy above?
>
> (d) Can you design a better strategy? What's its winning probability?
>
> [Hint: What is "obvious" is not obvious.]
>
> (d) Can you design a better strategy?
>
> [Hint: What is "obvious" is not obvious.]

Official Answer

Source: physical PDF page 14 (2019-final.pdf).

We pick door 3 first, then always switch. The winning probability is $p(E) = p(W = 1, M = 2) + p(W = 2, M = 1) = 0.5 \cdot 1 + 0.3 \cdot 1 = 0.8$.

Supplementary Explanation

Answer. Initially choose door 3 and always switch to the remaining unopened door.

Derivation/proof. Door 3 initially has the prize with probability 0.2. If that initial choice is wrong, Monty must reveal the other losing unchosen door, leaving the prize behind the remaining unopened door. Switching then wins. If the initial choice is right, switching loses. Thus this strategy wins exactly when $W \neq 3$, an event of probability $0.5 + 0.3 = 0.8$, which exceeds the previous 0.55 and proves it is better.

Since $0.8 > 0.55$, the strategy is strictly better as requested; no separate claim of global optimality is needed.

To maximize the probability of success by always switching, initially choose the door with the smallest prior probability of containing the prize. Switching succeeds exactly when the initial choice was wrong.

P029

Random Variables

2025 Fall Midterm | Question 10(a)(i) | Source page 21 | Difficulty 1
Q-2025F-MID-P10-a-i-E | Answer provenance: official

> (a) [6 points, 2 pts each] Let $X_1, X_2, \ldots, X_{100}$ be the outcomes of 100 independent rolls of a fair die.
>
> (i) [2 points] What are $E(X_1)$?

Official Answer

Source: physical PDF page 22 (COMP2711_F2025_Midterm.pdf).

> (a) (i) For a fair die, each outcome X_1 is uniformly distributed over $\{1, 2, 3, 4, 5, 6\}$. The expected value is:
>
> $$E(X_1) = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$
>
> The variance is:
>
> $$\text{var}(X_1) = E(X_1^2) - [E(X_1)]^2$$
>
> where $E(X_1^2) = \frac{1^2+2^2+3^2+4^2+5^2+6^2}{6} = \frac{91}{6}$, so:
>
> $$\text{var}(X_1) = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - \frac{12.25}{1} = \frac{91}{6} - \frac{73.5}{6} = \frac{17.5}{6} \approx 2.9167.$$

Supplementary Explanation

Answer. $E(X_1) = 7/2$.

Derivation/proof. Each value j=1,...,6 has probability 1/6. The discrete expectation is $E(X_1) = \sum_{j=1}^{6} j/6 = (1+2+3+4+5+6)/6 = 21/6 = 7/2$. Alternatively pair one with six, two with five, and three with four: each pair sums to seven, giving mean 3.5. Independence of different rolls is not needed for this single-die mean.

P030

Random Variables

2019 Final | Question 12(b) | Source page 15 | Difficulty 1 | 2 marks
Q-2019-FINAL-P12-b | Answer provenance: official

> Suppose the time line is discretized into intervals, with the length of each interval being 1 second. A web server has probability p $(0 < p < 1)$ of getting 1 visit in each time interval and probability $1 - p$ of getting 0 visit.
>
> (b) How many visits is the server expected to get during 1 hour?

Official Answer

Source: physical PDF page 15 (2019-final.pdf).

1 hour has 3600 seconds. There are 3600 independent Bernoulli trials. So the expected number of visits is $3600p$.

Supplementary Explanation

Answer. $3600p$.

Derivation/proof. There are $60 \cdot 60 = 3600$ one-second intervals in an hour. Let I_t indicate a visit in second t. The total is $N = \sum_{t=1}^{3600} I_t$, with $E(I_t) = p$. Thus $E(N) = \sum p = 3600p$ by linearity. Independence is not needed for this mean calculation.

Even if all seconds share a single Bernoulli variable, the expected total remains $3600p$.

With I_t indicating an event at second t, linearity gives $E(N) = \sum_{t=1}^{3600} E(I_t) = 3600p$. Thus independence across seconds is not required for the expectation, even if a binomial distribution is not justified.

P031

Random Variables

2017 Final | Question 6(a) | Source page 10 | Difficulty 1 | 1 mark
Q-2017-FINAL-P06-a | Answer provenance: official

> 64 fair coins are flipped independently randomly and then arranged in an 8 by 8 block randomly without looking. Each row contains eight coins and there are eight rows. If all eight coins in a row have the same symbol (heads or tails), we call it a homogeneous row. Let X denote the number of homogeneous rows in the block.
>
> (a) If all coin flips come up heads, what is the value of X?

Official Answer

Source: physical PDF page 10 (2017-final.pdf).

8.

Supplementary Explanation

Answer. $X = 8$.

Derivation/proof. Each row consists of eight heads and therefore satisfies the definition of homogeneous. There are eight rows, so exactly eight row indicators equal one and their sum X is eight, not sixty-four.

In indicator notation, $I_1 = \cdots = I_8 = 1$ and $X = \sum_{i=1}^{8} I_i = 8$.

P032

Random Variables

2016 Final | Question 1(b) | Source page 3 | Difficulty 1 | 2 marks
Q-2016-FINAL-P01-b | Answer provenance: official

> You are given two fair dice where each dice has the number 1 to 6 printed on it. You are also given one wheel so that a random spin of the wheel returns an integer from 2 to 12 in a uniform distribution. Answer the following questions.
>
> (b) You are now playing a boardgame. Each time you move your token x steps forward if you spin the wheel and obtain a number x. What is the expected number of steps moved if you spin the wheel 5 times?

Official Answer

Source: PDF physical page 3 (2016-final.pdf).

(b) $E(5x) = 5E(x) = 5\frac{2+3+4+5+6+7+8+9+10+11+12}{11} = 35$

Supplementary Explanation

Answer. 35 steps.

Derivation/proof. For spin X_i, $E[X_i] = (2 + 3 + \cdots + 12)/11 = 7$. The total is $S = X_1 + \cdots + X_5$, so $E[S] = \sum_{i=1}^{5} E[X_i] = 5 \cdot 7 = 35$. Independence is not needed for this use of linearity.

The single-spin sum is 77, so its expectation is $77/11 = 7$. The total is a sum of five random variables, rather than five copies of the same realized spin.

P033

Random Variables

2016 Final | Question 2(a) | Source page 5 | Difficulty 1 | 3 marks
Q-2016-FINAL-P02-a | Answer provenance: official

> Assume that $f(x)$ is a linear function in the form of $f(x) = ax + b$ where a and b are real numbers. Let n be a random variable with mean m and variance v. Expressing the following in terms of a, b, m, v and some constants. Show your steps.
>
> (a) Variance of $f(n)$.

Official Answer

Source: PDF physical page 5 (2016-final.pdf).

Assume that S is the sample space of n.

(a) Variance of $f(n)$, first need to find $E(f(n))$. By the linearity of expectation, $E(f(n)) = am + b$.

$$\begin{aligned} Var(f(n)) &= \sum_{x:x\in S} p(x)(ax + b - E(f(n))^2 \\ &= \sum_{x:x\in S} p(x)(ax + b - am - b)^2 \\ &= \sum_{x:x\in S} p(x)a^2(x - m)^2 \\ &= a^2 \sum_{x:x\in S} p(x)(x - m)^2 = a^2 v. \end{aligned}$$

Supplementary Explanation

Answer. $\mathrm{Var}(f(n)) = a^2 v$.

Derivation/proof. Linearity gives $E[f(n)] = am + b$. Therefore $\mathrm{Var}(f(n)) = E[(an + b - (am + b))^2] = a^2 E[(n - m)^2] = a^2 v$. This also covers $a = 0$, when $f(n)$ is constant.

The centered variable is $a(n - m)$ because the shift cancels.

P034

Random Variables

2015 Final | Question 2(c) | Source page 5 | Difficulty 1 | 2 marks
Q-2015-FINAL-P02-c | Answer provenance: official

> For this problem, the time is discretized into equal-sized intervals and the length of each interval is 1 minute.
>
> An online store sells bags. At each time interval, the probability of one customer arriving is p, and the probability of no customer arriving is $1-p$. Each customer buys one bag with probability q and does not buy anything with probability $1-q$.
>
> For simplicity, assume that each customer arrives at the store, completes purchase, if any, and leaves the store within the 1-minute time interval.
>
> Let X be the total number of bags sold within 60 minutes.
>
> (c) What is the expectation $E(X)$ of X?
>
> There is no need to justify your answers in this problem.

Official Answer

Source: PDF physical page 5 (2015-final.pdf).

This is an independent Bernoulli trials process with success probability pq.

(c) $E(X) = 60pq$.

Supplementary Explanation

Answer. $E(X) = 60pq$.

Derivation/proof. Let I_t indicate a sale in minute t. Then $X = \sum_{t=1}^{60} I_t$ and $E(I_t) = P(I_t = 1) = pq$. Linearity gives $E(X) = \sum_{t=1}^{60} pq = 60pq$.

The multiplication formula gives $P(I_t = 1) = P(\text{arrival})P(\text{purchase} \mid \text{arrival}) = pq$; arrival and purchase need not be independent. The expectation calculation remains valid even if different minutes are dependent. The source solution invokes an independent-trials model for the other parts; that additional assumption is unnecessary here.

P035

Random Variables

2025 Fall Midterm | Question 10(a)(i) | Source page 21 | Difficulty 2
Q-2025F-MID-P10-a-i-var | Answer provenance: official

> (a) [6 points, 2 pts each] Let $X_1, X_2, \ldots, X_{100}$ be the outcomes of 100 independent rolls of a fair die.
>
> (i) [2 points] What are $var(X_1)$?

Official Answer

Source: physical PDF page 22 (COMP2711_F2025_Midterm.pdf).

> (a) (i) For a fair die, each outcome X_1 is uniformly distributed over $\{1, 2, 3, 4, 5, 6\}$. The expected value is:
>
> $$E(X_1) = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$
>
> The variance is:
>
> $$\text{var}(X_1) = E(X_1^2) - [E(X_1)]^2$$
>
> where $E(X_1^2) = \frac{1^2+2^2+3^2+4^2+5^2+6^2}{6} = \frac{91}{6}$, so:
>
> $$\text{var}(X_1) = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - \frac{12.25}{1} = \frac{91}{6} - \frac{73.5}{6} = \frac{17.5}{6} \approx 2.9167.$$

Supplementary Explanation

Answer. $\text{var}(X_1) = 35/12$.

Derivation/proof. The second moment is $E(X_1^2) = (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2)/6 = 91/6$, and $E(X_1) = 21/6 = 7/2$. Hence $\text{var}(X_1) = E(X_1^2) - [E(X_1)]^2 = 91/6 - 49/4 = (182 - 147)/12 = 35/12$. This is positive, as expected for six distinct outcomes. The printed value $17.5/6$ is exactly the same number.

P036

Random Variables

2025 Fall Midterm | Question 10(a)(ii) | Source page 21 | Difficulty 2
Q-2025F-MID-P10-a-ii-E | Answer provenance: official

> (a) Let $X_1, X_2, \ldots, X_{100}$ be the outcomes of 100 independent rolls of a fair die. (ii) [2 points] Define the random variable Y to be $X_1 - 2X_2 + X_3$.
>
> What are $E(Y)$?

Official Answer

Source: physical PDF page 22 (COMP2711_F2025_Midterm.pdf).

> (iii) For $Y = X_1 - 2X_2 + X_3$, since X_1, X_2, X_3 are independent:
>
> $$E(Y) = E(X_1) - 2E(X_2) + E(X_3) = 3.5 - 2 \cdot 3.5 + 3.5 = 0.$$
>
> $$\operatorname{var}(Y) = \operatorname{var}(X_1) + (-2)^2 \operatorname{var}(X_2) + \operatorname{var}(X_3) = \frac{17.5}{6} + 4 \cdot \frac{17.5}{6} + \frac{17.5}{6} = \frac{105}{6} = 17.5.$$

Supplementary Explanation

Answer. $E(Y) = 0$.

Derivation/proof. Every die has mean $7/2$. By linearity, $E(Y) = E(X_1) - 2E(X_2) + E(X_3) = 7/2 - 2(7/2) + 7/2 = 0$. The coefficient sum $1 - 2 + 1 = 0$ explains the cancellation. Linearity holds even for dependent variables; the independence given here is needed for the variance calculation.

P037

Random Variables

2025 Fall Midterm | Question 10(a)(ii) | Source page 21 | Difficulty 2
Q-2025F-MID-P10-a-ii-var | Answer provenance: official

> (a) Let $X_1, X_2, \ldots, X_{100}$ be the outcomes of 100 independent rolls of a fair die. (ii) [2 points] Define the random variable Y to be $X_1 - 2X_2 + X_3$.
>
> What are $var(Y)$?

Official Answer

Source: physical PDF page 22 (COMP2711_F2025_Midterm.pdf).

> (iii) For $Y = X_1 - 2X_2 + X_3$, since X_1, X_2, X_3 are independent:
>
> $$E(Y) = E(X_1) - 2E(X_2) + E(X_3) = 3.5 - 2 \cdot 3.5 + 3.5 = 0.$$
>
> $$\text{var}(Y) = \text{var}(X_1) + (-2)^2 \,\text{var}(X_2) + \text{var}(X_3) = \frac{17.5}{6} + 4 \cdot \frac{17.5}{6} + \frac{17.5}{6} = \frac{105}{6} = 17.5.$$

Supplementary Explanation

Answer. $\text{var}(Y) = 35/2$.

Derivation/proof. Each die has variance $35/12$. Independence makes all covariance terms zero. Thus $\text{var}(Y) = 1^2 \,\text{var}(X_1) + (-2)^2 \,\text{var}(X_2) + 1^2 \,\text{var}(X_3) = (1 + 4 + 1)35/12 = 210/12 = 35/2$. This is 17.5, not $35/6$, the variance of the extra two-die difference X appearing on the answer page. Match the answer by the defined variable rather than the shifted numbering.

P038

Random Variables

2025 Fall Midterm | Question 10(a)(iii) | Source page 21 | Difficulty 2
Q-2025F-MID-P10-a-iii-E | Answer provenance: official

> (a) Let $X_1, X_2, \ldots, X_{100}$ be the outcomes of 100 independent rolls of a fair die. (iii) [2 points] Define the random variable $Z = X_1 - X_2 + X_3 - X_4 + \cdots + X_{99} - X_{100}$.
>
> What are $E(Z)$?

Official Answer

Source: physical PDF page 22 (COMP2711_F2025_Midterm.pdf).

> (iv) For $Z = X_1 - X_2 + X_3 - \cdots + X_{99} - X_{100}$, there are 50 positive terms and 50 negative terms:
>
> $$E(Z) = \sum_{i=1}^{50} E(X_{2i-1}) - \sum_{i=1}^{50} E(X_{2i}) = 50 \cdot 3.5 - 50 \cdot 3.5 = 0.$$
>
> $$\operatorname{var}(Z) = \sum_{i=1}^{50} \operatorname{var}(X_{2i-1}) + \sum_{i=1}^{50} \operatorname{var}(X_{2i}) = 50 \cdot \frac{17.5}{6} + 50 \cdot \frac{17.5}{6} = 100 \cdot \frac{17.5}{6} = \frac{1750}{6}.$$

Supplementary Explanation

Answer. $E(Z) = 0$.

Derivation/proof. Linearity gives $E(Z) = E(X_1) - E(X_2) + \cdots + E(X_{99}) - E(X_{100})$. There are fifty positive and fifty negative terms, each with mean $7/2$, so $E(Z) = 50(7/2) - 50(7/2) = 0$. Equivalently, sum fifty adjacent differences $E(X_{2j-1} - X_{2j})$, each with mean zero.

P039

Random Variables

2025 Fall Midterm | Question 10(a)(iii) | Source page 21 | Difficulty 2
Q-2025F-MID-P10-a-iii-var | Answer provenance: official

> (a) Let $X_1, X_2, \ldots, X_{100}$ be the outcomes of 100 independent rolls of a fair die. (iii) [2 points] Define the random variable $Z = X_1 - X_2 + X_3 - X_4 + \cdots + X_{99} - X_{100}$.
>
> What are $var(Z)$?

Official Answer

Source: physical PDF page 22 (COMP2711_F2025_Midterm.pdf).

> (iv) For $Z = X_1 - X_2 + X_3 - \cdots + X_{99} - X_{100}$, there are 50 positive terms and 50 negative terms:
>
> $$E(Z) = \sum_{i=1}^{50} E(X_{2i-1}) - \sum_{i=1}^{50} E(X_{2i}) = 50 \cdot 3.5 - 50 \cdot 3.5 = 0.$$
>
> $$\text{var}(Z) = \sum_{i=1}^{50} \text{var}(X_{2i-1}) + \sum_{i=1}^{50} \text{var}(X_{2i}) = 50 \cdot \frac{17.5}{6} + 50 \cdot \frac{17.5}{6} = 100 \cdot \frac{17.5}{6} = \frac{1750}{6}.$$

Supplementary Explanation

Answer. $\text{var}(Z) = 875/3$.

Derivation/proof. The hundred die variables are mutually independent and each has variance $35/12$. Every coefficient is $+1$ or -1, so every squared coefficient is one and all cross-covariance terms vanish. Consequently $\text{var}(Z) = 100 \cdot 35/12 = 3500/12 = 875/3$. The printed $1750/6$ reduces to the same $875/3$. Although the mean is zero, all hundred variance contributions accumulate.

P040

Random Variables

2023 Fall Midterm | Question 9(a) | Source page 19 | Difficulty 2
Q-2023F-MID-P09-a-expectation | Answer provenance: official

> Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),
>
> $$p(X=i)=\begin{cases}\frac{2}{3} & \text{if } i=1,\\ \frac{1}{3} & \text{if } i=-1,\\ 0 & \text{otherwise.}\end{cases}$$
>
> For (d): This robot is standing at position 0. It stops once it reaches to position $+1$ cm. Before stopping, the robot moves T times where T is a random variable.
>
> (a) Calculate $E(X)$.

Official Answer

Source: physical PDF page 19 (COMP2711_F2023_Midterm.pdf).

> (a)
>
> $$\begin{aligned}E(X)&=1\cdot\frac{2}{3}-1\cdot\frac{1}{3}\\&=\frac{1}{3}\\E(X^2)&=1\cdot\frac{2}{3}+1\cdot\frac{1}{3}\\&=1\\V(X)&=E(X^2)-E(X)^2=\frac{8}{9}.\end{aligned}$$

Supplementary Explanation

Answer. $E(X)=1/3$.

Derivation/proof. The atomic target is $E(X)$. Using the displayed two-point PMF, $E(X)=1\cdot(2/3)+(-1)\cdot(1/3)=1/3$ cm. The probabilities sum to 1, and the result lies between the two possible displacements −1 and 1. This is the signed average displacement per move; it does not require any assumption about dependence between different moves.

No independence assumption between different steps is needed. The positive mean agrees with the larger rightward probability.

P041

Random Variables

2023 Fall Midterm | Question 9(a) | Source page 19 | Difficulty 2
Q-2023F-MID-P09-a-variance | Answer provenance: official

> Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),
>
> $$p(X=i)=\begin{cases}\frac{2}{3} & \text{if } i=1,\\ \frac{1}{3} & \text{if } i=-1,\\ 0 & \text{otherwise.}\end{cases}$$
>
> For (d): This robot is standing at position 0. It stops once it reaches to position $+1$ cm. Before stopping, the robot moves T times where T is a random variable.
>
> (a) Calculate $V(X)$.

Official Answer

Source: physical PDF page 19 (COMP2711_F2023_Midterm.pdf).

> (a)
>
> $$\begin{aligned}E(X)&=1\cdot\frac{2}{3}-1\cdot\frac{1}{3}\\&=\frac{1}{3}\\E(X^2)&=1\cdot\frac{2}{3}+1\cdot\frac{1}{3}\\&=1\\V(X)&=E(X^2)-E(X)^2=\frac{8}{9}.\end{aligned}$$

Supplementary Explanation

Answer. $V(X)=8/9$.

Derivation/proof. The atomic target is $V(X)$. Directly, $E(X)=2/3-1/3=1/3$ and $E(X^2)=1^2(2/3)+(-1)^2(1/3)=1$. Thus $V(X)=E(X^2)-E(X)^2=1-1/9=8/9$ square cm. As a separate check, the centered-square calculation is $(1-1/3)^2(2/3)+(-1-1/3)^2(1/3)=8/27+16/27=8/9$.

The independent centered-square check is $\frac{8}{27}+\frac{16}{27}=\frac{8}{9}$.

P042

Random Variables

2023 Fall Midterm | Question 9(b) | Source page 19 | Difficulty 2
Q-2023F-MID-P09-b-expectation | Answer provenance: official

> Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),
>
> $$p(X=i)=\begin{cases}\frac{2}{3} & \text{if } i=1,\\ \frac{1}{3} & \text{if } i=-1,\\ 0 & \text{otherwise.}\end{cases}$$
>
> For (d): This robot is standing at position 0. It stops once it reaches to position $+1$ cm. Before stopping, the robot moves T times where T is a random variable.
>
> (b) If $Y=X-1$, calculate $E(Y)$.

Official Answer

Source: physical PDF page 19 (COMP2711_F2023_Midterm.pdf).

> (b)
>
> $$E(Y)=E(X-1)=E(X)-1=-\frac{2}{3}$$
>
> $$V(Y)=V(X-1)=V(X)=\frac{8}{9}$$

Supplementary Explanation

Answer. $E(Y)=-2/3$.

Derivation/proof. The atomic target is $E(Y)$. If $X=1$, then $Y=0$ with probability $2/3$; if $X=-1$, then $Y=-2$ with probability $1/3$. Hence $E(Y)=0(2/3)+(-2)(1/3)=-2/3$. Equivalently, recomputing $E(X)=1/3$ from the PMF gives $E(X-1)=E(X)-1=-2/3$. Both computations use only the one-step distribution.

The mean lies in the support interval $[-2,0]$; a negative mean is not a negative probability.

P043

Random Variables

2023 Fall Midterm | Question 9(b) | Source page 19 | Difficulty 2
Q-2023F-MID-P09-b-variance | Answer provenance: official

> Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),
>
> $$p(X=i)=\begin{cases}\frac{2}{3} & \text{if } i=1,\\ \frac{1}{3} & \text{if } i=-1,\\ 0 & \text{otherwise.}\end{cases}$$
>
> For (d): This robot is standing at position 0. It stops once it reaches to position $+1$ cm. Before stopping, the robot moves T times where T is a random variable.
>
> (b) If $Y=X-1$, calculate $V(Y)$.

Official Answer

Source: physical PDF page 19 (COMP2711_F2023_Midterm.pdf).

> (b)
>
> $$E(Y)=E(X-1)=E(X)-1=-\frac{2}{3}$$
>
> $$V(Y)=V(X-1)=V(X)=\frac{8}{9}$$

Supplementary Explanation

Answer. $V(Y)=8/9$.

Derivation/proof. The atomic target is $V(Y)$. Its values are 0 and -2 with probabilities $2/3$ and $1/3$, so $E(Y)=-2/3$ and $E(Y^2)=4/3$. Therefore $V(Y)=4/3-(-2/3)^2=12/9-4/9=8/9$. In general $(X-1)-E(X-1)=X-E(X)$, proving directly why subtracting a constant cannot change variance.

From the new distribution, $4/3-4/9=8/9$.

P044

Random Variables

2023 Fall Midterm | Question 9(c) | Source page 19 | Difficulty 2 | 2 marks
Q-2023F-MID-P09-c | Answer provenance: official

> Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),
>
> $$p(X=i)=\begin{cases}\frac{2}{3} & \text{if } i=1,\\ \frac{1}{3} & \text{if } i=-1,\\ 0 & \text{otherwise.}\end{cases}$$
>
> For (d): This robot is standing at position 0. It stops once it reaches to position $+1$ cm. Before stopping, the robot moves T times where T is a random variable.
>
> (c) If $Z=X(X-1)$, calculate $E(Z)$.

Official Answer

Source: physical PDF page 19 (COMP2711_F2023_Midterm.pdf).

> (c) $E(Z)=E(X(X-1))=E(X^2-X)=E(X^2)-E(X)=\frac{2}{3}$

Supplementary Explanation

Answer. $E(Z)=2/3$.

Derivation/proof. When $X=1$, $Z=1(1-1)=0$; when $X=-1$, $Z=(-1)(-2)=2$. Hence $E(Z)=0(2/3)+2(1/3)=2/3$. Independently, $Z=X^2-X$ and linearity gives $E(Z)=E(X^2)-E(X)=1-1/3=2/3$. There is no independence between X and X−1: both are functions of the same outcome, so multiplying their expectations would be invalid.

Explicitly $E(Z)=1-1/3=2/3$; there is no independence between the two factors.

P045

Random Variables

2023 Fall Midterm | Question 9(d)(i) | Source page 19 | Difficulty 2 | 1 mark
Q-2023F-MID-P09-d-i | Answer provenance: official

> Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),
>
> $$p(X=i)=\begin{cases}\frac{2}{3} & \text{if } i=1,\\ \frac{1}{3} & \text{if } i=-1,\\ 0 & \text{otherwise.}\end{cases}$$
>
> For (d): This robot is standing at position 0. It stops once it reaches to position $+1$ cm. Before stopping, the robot moves T times where T is a random variable.
>
> (i) When k is a positive integer, calculate $P(T=2k)$.

Official Answer

Source: physical PDF page 20 (COMP2711_F2023_Midterm.pdf).

> (d) Denote every right move to be R and every left move to be L. Since the robot will stop once it is in position +1 cm, the last move can only be from 0 to +1 cm. Thus, the last move is R, and before the last move, we need to have the same number of R and L. Denote the subsequence excluding the last move to be sequence S. L is legal if when we read from left to right, we have always encountered at least as many L as R.
>
> (i) Since S has the same number of R and L, the length of S must be even. Adding the last move R, it is impossible for T to be an even number. Thus $P(T=2k)=0$.

Supplementary Explanation

Answer. $P(T=2k)=0$ for every positive integer k.

Derivation/proof. After t moves, the position is the sum of t numbers each equal to +1 or −1. Both possible increments are congruent to 1 modulo 2, so the position is congruent to t modulo 2. Reaching +1 therefore requires odd t. In particular, no trajectory can first reach +1 at time 2k, and the event is empty. Its probability is zero under any dependence structure of the successive moves.

The event $\{T=2k\}$ is empty, so its probability must be zero. The proof does not use independence between different moves or eventual arrival; trajectories that never arrive cannot create an even hitting time.

P046

Random Variables

2022 Final Part 2 | Question 6(a) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P06-a-expectation | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the expectation of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (a) X_e is the indicator random variable that indicates the event of a pair of vertices $e = (u, v)$ having an edge. X_e is a Bernoulli trial, thus $E(X_e) = p$, $V(X_e) = p(1-p)$.
>
> There are $\binom{n}{2}$ pairs of vertices in total. Thus $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $E[X] = \binom{n}{2}p$.

Derivation/proof. Index the unordered pairs by e and define I_e=1 for an existing edge and 0 otherwise. The edge count is X=∑I_e. Each I_e has mean p. Summing by linearity over N=n(n−1)/2 pairs gives E[X]=Np. This matches the binomial mean for X distributed as Binomial(N,p); linearity alone already suffices for this calculation.

Write $X = \sum_e I_e$. Since $E[I_e] = 1 \cdot p + 0 \cdot (1-p) = p$,

$$E[X] = \sum_e p = Np = \binom{n}{2}p.$$

Linearity itself does not require independence.

P047

Random Variables

2022 Final Part 2 | Question 6(a) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P06-a-variance | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the variance of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (a) X_e is the indicator random variable that indicates the event of a pair of vertices $e = (u, v)$ having an edge. X_e is a Bernoulli trial, thus $E(X_e) = p$, $V(X_e) = p(1-p)$.
>
> There are $\binom{n}{2}$ pairs of vertices in total. Thus $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $\mathrm{Var}(X) = \binom{n}{2}p(1-p)$.

Derivation/proof. For one edge indicator I, E[I²]=E[I]=p, so Var(I)=p−p². In the stated random graph distinct edge indicators are independent, giving covariance zero for every distinct pair. The variance of their sum is therefore N(p−p²), with N=C(n,2). At p=0 or1 the answer is zero, correctly reflecting a deterministic graph.

Because $I_e^2 = I_e$, its variance is $p - p^2 = p(1-p)$. Independence makes the covariance zero for each distinct pair; hence

$$\mathrm{Var}(X) = \sum_e \mathrm{Var}(I_e) = \binom{n}{2} p(1-p).$$

Equal marginal probabilities alone would not justify dropping covariance terms.

P048

Random Variables

2020 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P05-b-expectation | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the expectation of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

X_e is the indicator random variable that indicates the event, pair of vertices $e = (u, v)$ has an edge. X_e is a Bernoulli trial, thus $E(X_e) = p, V(X_e) = p(1-p)$. There are $\binom{n}{2}$ pairs of vertices in total. So that $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $E[X] = \binom{n}{2}p$.

Derivation/proof. For each of the $\binom{n}{2}$ unordered vertex pairs e let I_e be 1 when its edge exists and 0 otherwise. Then $X = \sum_e I_e$ and $E[I_e] = 1 \cdot p + 0 \cdot (1-p) = p$. By linearity, $E[X] = \sum_e p = \binom{n}{2}p$. Independence is not needed for this expectation step.

There is one indicator for each potential unordered edge. Therefore $E(X) = \sum_e E(I_e) = \binom{n}{2}p$ by linearity, without any need for independence in this particular step.

P049

Random Variables

2020 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P05-b-variance | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the variance of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

X_e is the indicator random variable that indicates the event, pair of vertices $e = (u, v)$ has an edge. X_e is a Bernoulli trial, thus $E(X_e) = p, V(X_e) = p(1-p)$. There are $\binom{n}{2}$ pairs of vertices in total. So that $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $\mathrm{Var}(X) = \binom{n}{2}p(1-p)$.

Derivation/proof. Write X as the sum of independent edge indicators I_e. Because $I_e^2 = I_e$, each has variance $E[I_e^2] - E[I_e]^2 = p - p^2 = p(1-p)$. Distinct indicators are independent by the graph model, so their covariances vanish. Adding $\binom{n}{2}$ identical variance terms gives the result.

P050

Random Variables

2018 Final | Question 8(a) | Source page 11 | Difficulty 2
Q-2018-FINAL-P08-a | Answer provenance: official

> Consider throwing m balls into n boxes at random. The probability that a ball ends up in any given box is $\frac{1}{n}$. After all balls have been thrown, let X be the number of balls in the first box and Y be the number of boxes that are empty.
>
> Show the steps of your calculations.
>
> a) What is $E(X)$? Show the steps of your calculations.

Official Answer

Source: physical PDF page 11 (2018-final.pdf).

$E(X)$: Each ball assignment is a Bernoulli trial with probability $1/n$ of success, and there are m trials. Therefore, X follows the Binomial distribution for which we know $E(X) = m(1/n) = m/n$. Alternative solution: Let X_i be the indicator random variable such that $X_i = 1$ if the i-th ball is in the first box, $X_i = 0$ otherwise. Obviously, $X = \sum_{i=1}^{m} X_i$. For $1 \leq i \leq m$, $E(X_i) = P(\text{ball } i \text{ in the first box}) = 1/n$. Therefore,

$$E(X) = E\left(\sum_{i=1}^{m} X_i\right) = \sum_{i=1}^{m} E(X_i) = \sum_{i=1}^{m} \frac{1}{n} = \frac{m}{n}.$$

Supplementary Explanation

Answer. $E(X) = m/n$.

Derivation/proof. Let I_i be one if ball i lands in box 1 and zero otherwise. Then $X = \sum_{i=1}^{m} I_i$ and $E(I_i) = P(I_i = 1) = 1/n$. Linearity gives $E(X) = \sum_{i=1}^{m} 1/n = m/n$. Equal marginals alone suffice for this particular expectation.

Even if every ball shares one uniformly selected box, the same expectation holds. This illustrates why independence must not be added to this proof.

Let I_i indicate that ball i enters the specified box. Then $X = \sum_{i=1}^{m} I_i$ and $E(I_i) = 1/n$, so linearity gives $E(X) = m/n$; independence between balls is not required for this expectation.

P051

Random Variables

2017 Final | Question 6(b) | Source page 10 | Difficulty 2 | 2 marks
Q-2017-FINAL-P06-b | Answer provenance: official

> 64 fair coins are flipped independently randomly and then arranged in an 8 by 8 block randomly without looking. Each row contains eight coins and there are eight rows. If all eight coins in a row have the same symbol (heads or tails), we call it a homogeneous row. Let X denote the number of homogeneous rows in the block.
>
> (b) Calculate $p(X = 1)$.

Official Answer

Source: physical PDF page 10 (2017-final.pdf).

For any given row, the chance it is homogeneous is $(1/2)^8 \cdot 2 = (1/2)^7$ (all heads or all tails). The chance that it is not homogeneous is $(1 - (1/2)^7)$. Overall, seven rows must not be homogeneous and one row must be homogeneous, so $p(X = 1) = 8(1/2)^7(1 - (1/2)^7)^7$.

Supplementary Explanation

Answer. $8(1/128)(127/128)^7$.

Derivation/proof. A given row is homogeneous in two of its 2^8 equally likely configurations, so its probability is $q = 2/2^8 = 1/128$. Distinct rows use disjoint independent coins. Choose which of the eight rows succeeds, multiply by q for it and $(1 - q)^7$ for the failures. The eight choices are disjoint, giving $8q(1 - q)^7$.

Randomly arranging the coins without observing their faces preserves the independent fair distribution at all sixty-four positions.

A row is homogeneous with probability $q = 2(1/2)^8 = 1/128$, since either all eight coins are heads or all are tails. The independent eight rows then give $P(X = 1) = 8(1/128)(127/128)^7$ by selecting its unique homogeneous row.

P052

Random Variables

2017 Final | Question 6(c) | Source page 10 | Difficulty 2 | 2 marks
Q-2017-FINAL-P06-c | Answer provenance: official

> 64 fair coins are flipped independently randomly and then arranged in an 8 by 8 block randomly without looking. Each row contains eight coins and there are eight rows. If all eight coins in a row have the same symbol (heads or tails), we call it a homogeneous row. Let X denote the number of homogeneous rows in the block.
>
> (c) Calculate $E(X)$.

Official Answer

Source: physical PDF page 10 (2017-final.pdf).

Consider the homogeneity of each row as its own indicator variable X_i, all X_i are mutually independent. So $E(X) = 8E(X_1) = 8(1/2)^7$.

Supplementary Explanation

Answer. $E(X) = 8/128 = 1/16$.

Derivation/proof. Let I_i indicate that row i is homogeneous. Then $X = \sum_{i=1}^{8} I_i$ and $E(I_i) = P(I_i = 1) = 2(1/2)^8 = 1/128$. By linearity, $E(X) = \sum E(I_i) = 8/128 = 1/16$. Independence is not needed for this expectation step.

The row probability uses the explicitly stated independence of the individual coin flips, even though linearity of expectation itself does not need row independence.

P053

Random Variables

2017 Final | Question 6(d) | Source page 10 | Difficulty 2 | 2 marks
Q-2017-FINAL-P06-d | Answer provenance: official

> 64 fair coins are flipped independently randomly and then arranged in an 8 by 8 block randomly without looking. Each row contains eight coins and there are eight rows. If all eight coins in a row have the same symbol (heads or tails), we call it a homogeneous row. Let X denote the number of homogeneous rows in the block.
>
> (d) Calculate $V(X)$.

Official Answer

Source: physical PDF page 10 (2017-final.pdf).

As above, $V(X) = 8V(X_1) = 8(1/2)^7(1 - (1/2)^7)$.

Supplementary Explanation

Answer. $V(X) = 8\frac{1}{128}\frac{127}{128} = \frac{127}{2048}$.

Derivation/proof. For row indicator I_i, $I_i^2 = I_i$, so $V(I_i) = E(I_i) - E(I_i)^2 = q(1-q)$ with $q = 1/128$. Disjoint row coin sets are independent, making all cross covariances zero. Therefore $V(X) = \sum_{i=1}^{8} V(I_i) = 8q(1-q) = 127/2048$.

Explicitly, $\mathrm{Cov}(I_i, I_j) = 0$ for $i \neq j$.

P054

Random Variables

2016 Final | Question 2(b) | Source page 5 | Difficulty 2 | 3 marks
Q-2016-FINAL-P02-b | Answer provenance: official

> Assume that $f(x)$ is a linear function in the form of $f(x) = ax + b$ where a and b are real numbers. Let n be a random variable with mean m and variance v. Expressing the following in terms of a, b, m, v and some constants. Show your steps.
>
> (b) Expectation of $f(n^2)$.

Official Answer

Source: PDF physical page 5 (2016-final.pdf).

Assume that S is the sample space of n.

(b) Expectation of $f(n^2)$:

$$\begin{aligned} E(f(n^2)) &= \sum_{x:x\in S} p(x)(ax^2+b) \\ &= a\sum_{x:x\in S} p(x)\cdot x^2 + \sum_{x:x\in S} p(x)\cdot b \\ &= a\sum_{x:x\in S} p(x)\cdot x^2 + b \end{aligned}$$

$$v = Var(n) = \left(\sum_{x:x\in S} p(x)x^2\right) - m^2$$

$$E(f(n^2)) = a(v+m^2)+b$$

Supplementary Explanation

Answer. $E[f(n^2)] = a(v+m^2)+b$.

Derivation/proof. From $v = E[n^2] - (E[n])^2$, obtain $E[n^2] = v + m^2$. Since $f(n^2) = an^2 + b$, linearity yields $E[f(n^2)] = aE[n^2] + b = a(v+m^2)+b$.

P055

Random Variables

2016 Final | Question 2(c) | Source page 5 | Difficulty 2 | 3 marks
Q-2016-FINAL-P02-c | Answer provenance: official

> Assume that $f(x)$ is a linear function in the form of $f(x) = ax + b$ where a and b are real numbers. Let n be a random variable with mean m and variance v. Expressing the following in terms of a, b, m, v and some constants. Show your steps.
>
> (c) Expectation of $f(n)^2$.

Official Answer

Source: PDF physical page 6 (2016-final.pdf).

(c) Expectation of $f(n)^2$:

$$\begin{aligned} E(f(n)^2) &= \sum_{x:x\in S} p(x)(ax+b)^2 \\ &= \sum_{x:x\in S} p(x)(a^2x^2 + 2abx + b^2) \\ &= a^2(v+m^2) + 2abm + b^2 \end{aligned}$$

Supplementary Explanation

Answer. $E[f(n)^2] = a^2(v+m^2) + 2abm + b^2$.

Derivation/proof. Expand $f(n)^2 = a^2n^2 + 2abn + b^2$. Taking expectations and using $E[n] = m$, $E[n^2] = v + m^2$ gives the result. As a check, subtract $(am+b)^2$ to recover a^2v, the variance from part (a).

P056

Random Variables

2016 Final | Question 2(d) | Source page 5 | Difficulty 2 | 3 marks
Q-2016-FINAL-P02-d | Answer provenance: official

> Assume that $f(x)$ is a linear function in the form of $f(x) = ax + b$ where a and b are real numbers. Let n be a random variable with mean m and variance v. Expressing the following in terms of a, b, m, v and some constants. Show your steps.
>
> (d) Expectation of $f(n)f(p)$, if p is another random variable that is independent of n and has the same distribution as n.

Official Answer

Source: PDF physical page 6 (2016-final.pdf).

(d) By expected product of independent random variables, mean of $f(n)f(p) = E(f(n))E(f(p))$. Thus,

$$\begin{aligned} E(f(n)f(p)) &= E(f(n))E(f(p)) \\ &= E(f(n))E(f(n)) \\ &= (am + b)^2 \end{aligned}$$

Supplementary Explanation

Answer. $E[f(n)f(p)] = (am + b)^2$.

Derivation/proof. Independence implies $E[np] = E[n]E[p] = m^2$. Expand $(an + b)(ap + b) = a^2np + abn + abp + b^2$ and take expectations: $a^2m^2 + 2abm + b^2 = (am + b)^2$. Equivalently, $f(n)$ and $f(p)$ are independent and each has expectation $am + b$.

Independence and equal means give $E(np) = m^2$, so the expanded expectation is $a^2m^2 + abm + abm + b^2 = (am + b)^2$.

P057

Random Variables

2021 Final Part 2 | Question Bonus(a) | Source page 4 | Difficulty 3 | 2 marks
Q-2021-FINAL2-BONUS-a | Answer provenance: official

> Let G be an undirected complete graph having n vertices. For a vertex v, $N(v)$ denotes the set of neighbors of v in G. A random walk on G is the following process, which occurs in a sequence of discrete steps: starting at a vertex v_0, we proceed at the first step to a neighbor of v_0 chosen uniformly at random. This may be thought of as choosing a random edge incident on v_0 and walking along it to a vertex $v_1 \in N(v_0)$. At the second step, we proceed to a randomly chosen neighbor of v_1, and so on.
>
> For two given vertices u and v of G, what's the expected number of steps of random walk that begins at u and ends upon first reaching v?

Official Answer

Source: physical PDF page 4 (2021-final-part-2.pdf).

> (a) The probability of visiting v at the next step is $p = \frac{1}{n-1}$ if the current vertex is not v. So the answer is $1/p = n - 1$.

Supplementary Explanation

Answer. $n - 1$ for distinct u, v; 0 if the starting vertex already equals the target.

Derivation/proof. For n≥2 and current vertex different from v, one of its n−1 equally likely neighbors is v. Thus each nonabsorbed step has success probability 1/(n−1), independent of which non-target vertex was reached. Let h be the expected remaining steps. First-step conditioning gives h=1+((n−2)/(n−1))h, hence h=n−1. If u=v, the first hitting time is already zero before any step.

After k steps the non-hitting probability is $(1 - p)^k$, a geometric tail proving finite expectation. Symmetry makes the remaining expectation the same at all non-target vertices. For $n = 2$ the next step hits with certainty, giving expectation one.

The boundary case is zero if $u = v$. Otherwise $h = 1 + \frac{n-2}{n-1}h$, so $h = n - 1$.

P058

Random Variables

2023 Fall Final | Question 4(d) | Source page 7 | Difficulty 4 | 4 marks
Q-2023F-FINAL-P04-d | Answer provenance: official

> The CSE Department of HKUST has a total of n light bulbs and m switches. We know that each light bulb is connected to at least one switch. By flipping a switch, the status of every light bulb connected to it will change (so if it was OFF it will be turned ON and vice versa). Assume X_i is the random variable that shows the status of the ith light bulb. $X_i = 0$ means light bulb i is OFF and $X_i = 1$ means it is ON. Also, let Y be the random variable that shows the number of light bulbs that are ON in the department. Note that the initial status of every light bulb is arbitrary.
>
> Recall the CSE Department with its light bulbs and switches again. Prove that it is possible to have at least half of the light bulbs being turned ON.
>
> (Hint: You might want to use some facts from the previous parts. You are also allowed to use the fact in part (c) even if you could not prove it)

Official Answer

Source: physical PDF page 7 (COMP2711_F2023_Final.pdf).

> (d) If we flip every switch with 50% probability, then on expectation half of the light bulbs will be on. So based on (c) there exists a way for switches such that at least half of the light bulbs are on.

Supplementary Explanation

Answer. There exists a switch configuration with at least $n/2$ bulbs ON (equivalently at least $\lceil n/2 \rceil$).

Derivation/proof. Choose independently for each of the m switches whether to flip it, with probability $1/2$. This is a deliberately constructed uniform distribution on all 2^m configurations. For bulb i, select one incident switch; toggling just that choice pairs each configuration with another in which bulb i has the opposite final state. Thus $E(X_i) = 1/2$ regardless of its initial state. Since $Y = \sum_i X_i$, linearity gives $E(Y) = n/2$, even though the bulb indicators may depend on one another. If every configuration had $Y < n/2$, their finite uniform average would be strictly below $n/2$, contradicting $E(Y) = n/2$. Therefore some configuration has $Y \geq n/2$. Equivalently, apply the finite-average lower-witness argument to $K = -Y$; that argument is proved locally, without importing the quarantined claim (c).

The 2^m equally likely switch configurations admit a one-to-one pairing by toggling any chosen switch incident to a fixed bulb. Exactly one member of each pair leaves that bulb on. Linearity of expectation holds even when the bulb indicators are dependent. Apply part (c) to $K = -Y$ to obtain $Y \geq n/2$, hence at least $\lceil n/2 \rceil$ lit bulbs.

P059

Random Variables

2021 Final Part 2 | Question Bonus(b) | Source page 4 | Difficulty 4 | 8 marks
Q-2021-FINAL2-BONUS-b | Answer provenance: official

> Let G be an undirected complete graph having n vertices. For a vertex v, $N(v)$ denotes the set of neighbors of v in G. A random walk on G is the following process, which occurs in a sequence of discrete steps: starting at a vertex v_0, we proceed at the first step to a neighbor of v_0 chosen uniformly at random. This may be thought of as choosing a random edge incident on v_0 and walking along it to a vertex $v_1 \in N(v_0)$. At the second step, we proceed to a randomly chosen neighbor of v_1, and so on.
>
> Redo (a) for the case where G is a line graph of n vertices, and u is at one end and v is at the other end.

Official Answer

Source: physical PDF page 5 (2021-final-part-2.pdf).

> (b) Number of the vertices as $1, \ldots, n$. Let T_i denote the number of steps to go from i to n. We have $T_n = 0$, so $E[T_n] = 0$. The goal is to find $E[T_1]$.
>
> From vertex 1, we can only go right, so $T_1 = 1 + T_2$ and $E[T_1] = 1 + E[T_2]$. Rewrite it as $E[T_1] - E[T_2] = 1$. For any $1 < i < n$, we go left or right with equal probability, so $E[T_i] = 1 + \frac{1}{2}E[T_{i-1}] + \frac{1}{2}E[T_{i+1}]$. Rearrange it as $E[T_i] - E[T_{i+1}] = 2 + E[T_{i-1}] - E[T_i]$. Now we have
>
> $$\begin{aligned} E[T_1] - E[T_2] &= 1 \\ E[T_2] - E[T_3] &= 3 \\ &\cdots \\ E[T_{n-1}] - E[T_n] &= 2n - 3. \end{aligned}$$
>
> Summing up, we have $E[T_1] = E[T_n] + 1 + 3 + \cdots + 2n - 3 = (n-1)^2$.

Supplementary Explanation

Answer. $(n-1)^2$.

Derivation/proof. Label the path vertices 1,...,n and let h_i be expected steps to reach n. Then h_n=0 and h_1=1+h_2. For 1<i<n, h_i=1+(h_(i−1)+h_(i+1))/2. Define d_i=h_i−h_(i+1); rearrangement gives d_1=1 and d_i=d_(i−1)+2, hence d_i=2i−1. Telescoping yields h_1−h_n=1+3+...+(2n−3)=(n−1)². Expectations are finite: from any vertex there is a positive uniformly bounded chance to follow right steps to n within n−1 steps, giving a geometric block tail. For n=1 the answer is zero.

For $n \geq 2$, from any non-target vertex, following rightward steps reaches the target within $n-1$ steps with conditional probability at least $2^{-(n-1)} > 0$. Each block has this uniform success lower bound, so a geometric tail controls the hitting time. Explicitly,

$$h_1 = h_1 - h_n = \sum_{i=1}^{n-1} d_i = 1 + 3 + \cdots + (2n-3) = (n-1)^2.$$

For $n = 1$, both endpoints coincide and the expectation is zero.

The difference recurrence is $d_i = d_{i-1} + 2$, giving $d_i = 2i - 1$.

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